

StEm65, Sexten, 20.07.23

Exploring the IGM with the Sherwood-Relics simulations

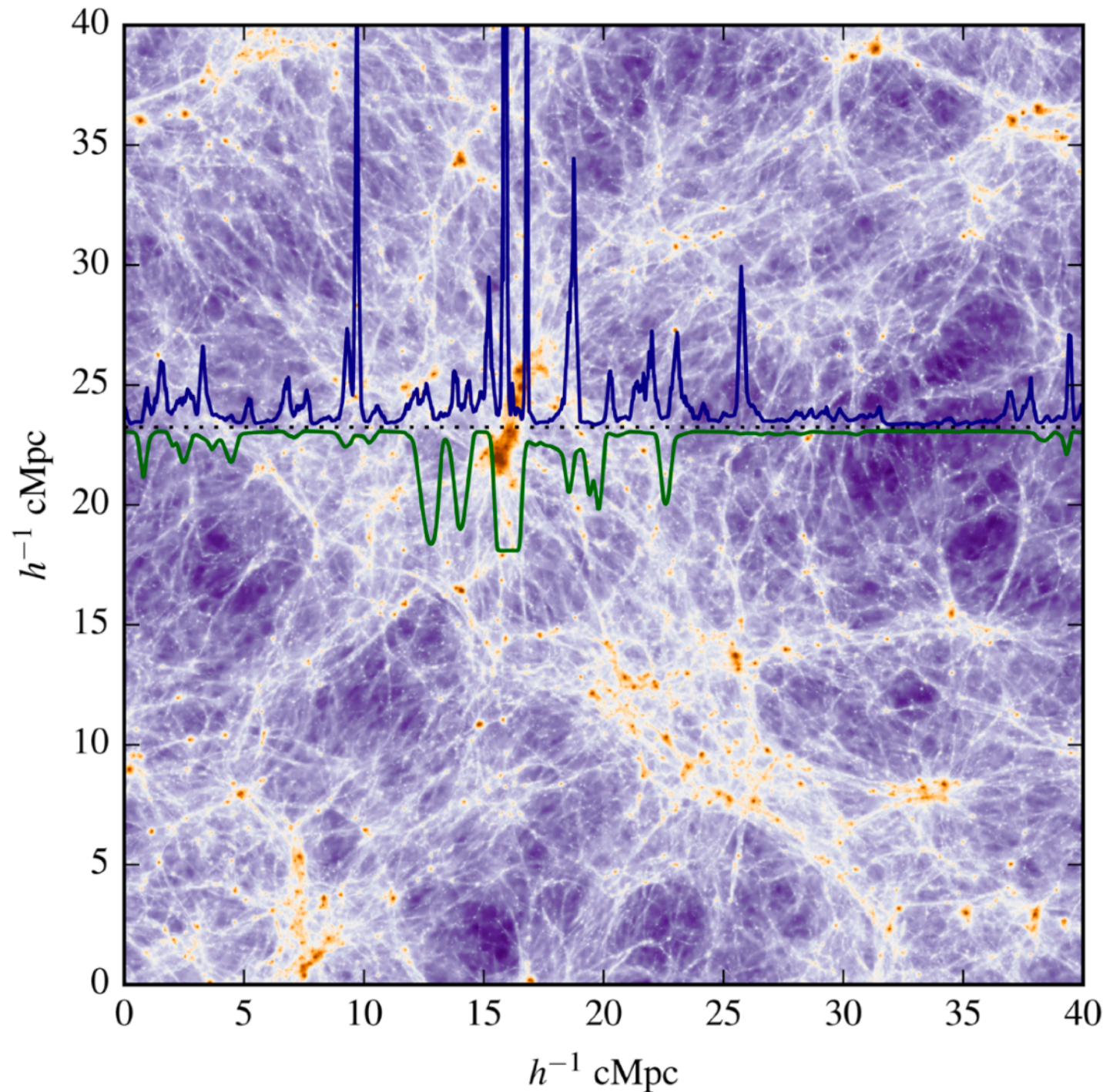
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Martin Haehnelt, Vid Iršič, Laura Keating,
Girish Kulkarni, **Margherita Molaro**,
Ewald Puchwein, Tomáš Šoltinský,
Matteo Viel

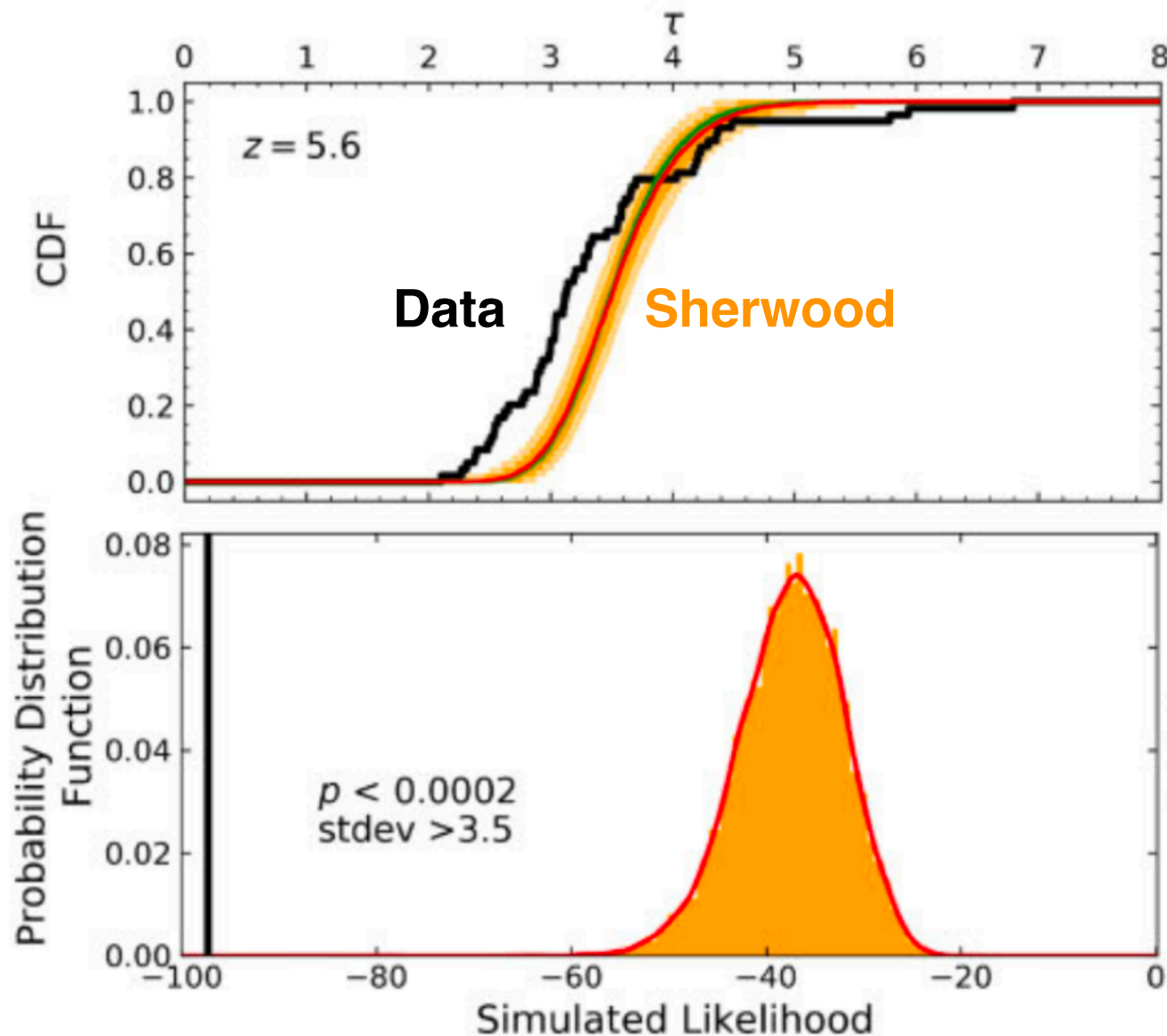
The Sherwood simulation suite



Bolton et al. (2017)

- 40-160 Mpc/h with $M_{\text{gas}} \sim 10^5 M_{\odot}/h$; Planck cosmology; hydrodynamical IGM calculations with P-Gadget-3 (Springel 2005)
- CDM and WDM (3.5 keV thermal relic) runs, varying thermal histories;
- Haardt & Madau (2012) UVB, assumed spatially uniform.
- Stellar and AGN feedback (for selected models) follows Puchwein & Springel (2013).
- Data are fully public, 41 papers so far. Agrees very well with Ly α forest data at $2 < z < 5$, but...

A late end to reionisation at $z \sim 5.3$



- Scatter in Ly- α forest transmission inconsistent with spatially uniform UV background at $z \geq 5.3$ - evidence for very late reionisation (e.g. [Kulkarni+19](#), [Keating+20](#)).
- Ly- α forest models with uniform UVB (inc. Sherwood) fail to capture the width of the cumulative τ_{eff} distribution.

Bosman et al. (2022)
see also Becker+15, Eilers+18, Fan+06



Sherwood-Relics

Hybrid radiation-hydrodynamical models of the Lyman-alpha forest.

Looking for the original Sherwood? Click [here](#).



Science & Technology
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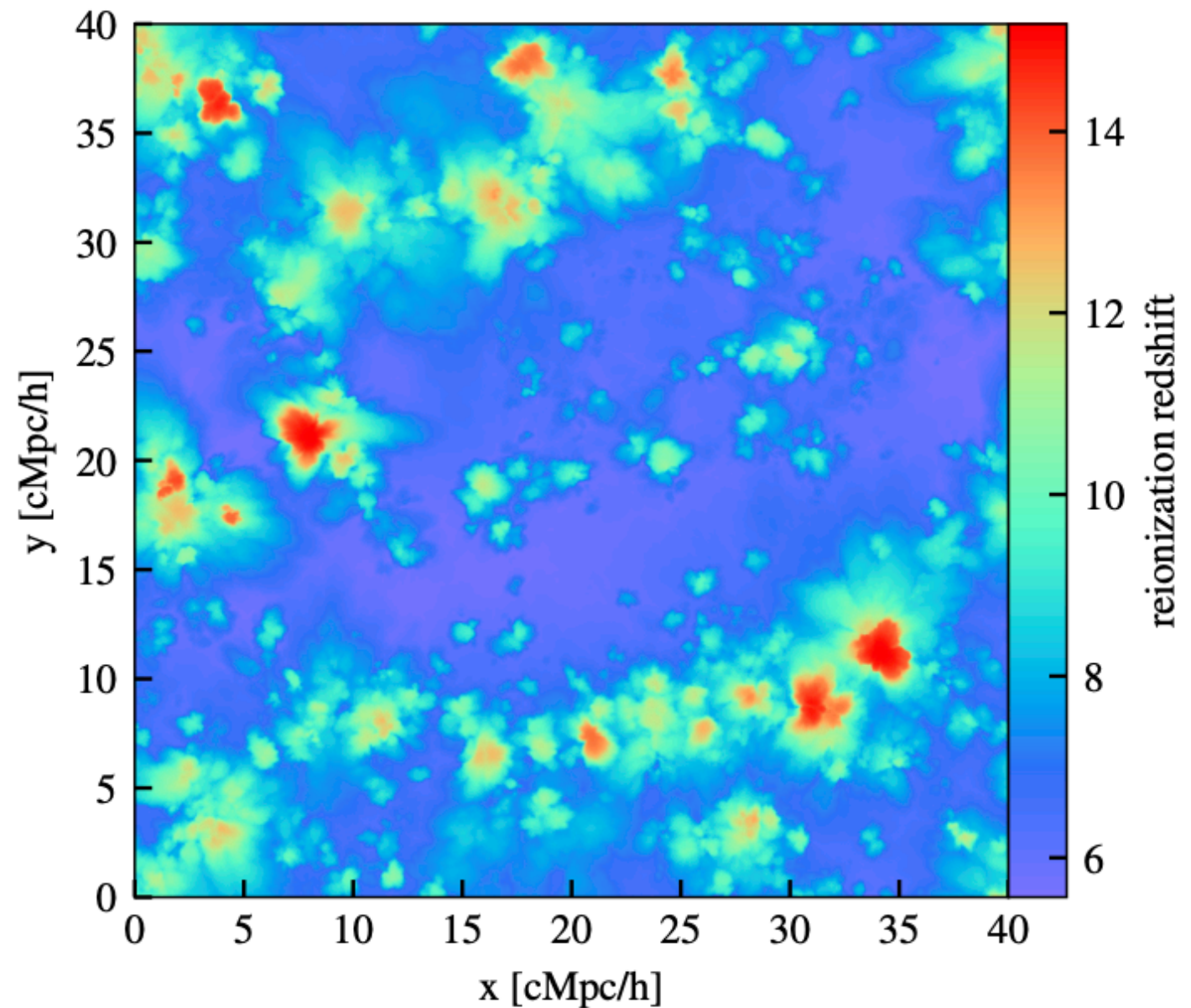


DiRAC

<http://www.nottingham.ac.uk/astronomy/sherwood-relics/>

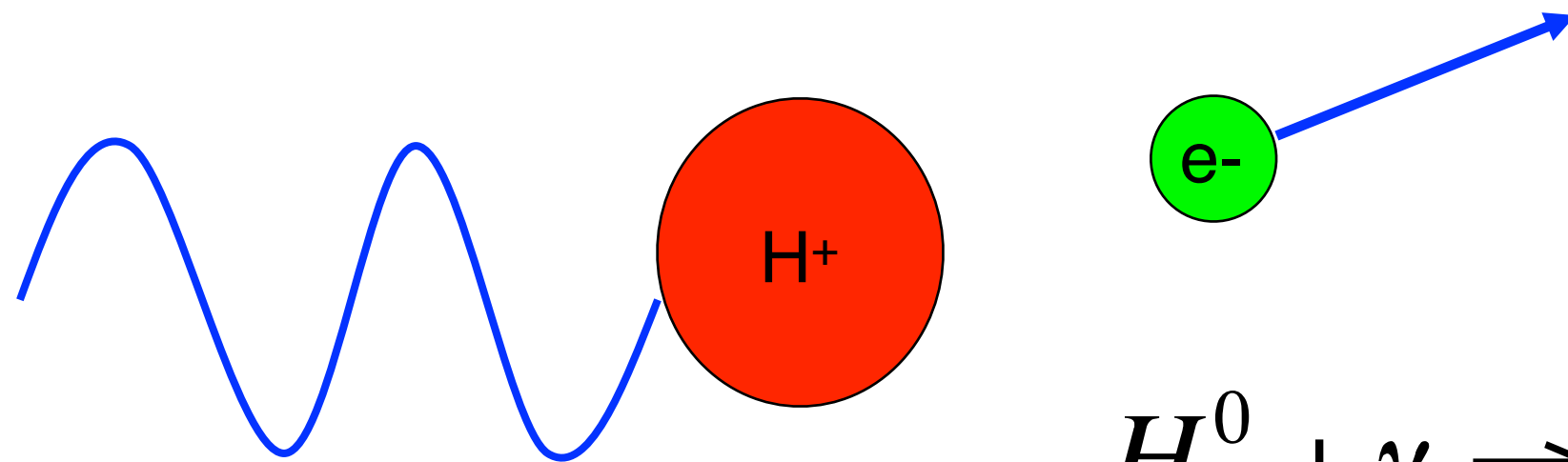
Technical improvements

- RT using GPU accelerated code ATON ([Aubert & Teyssier 2008](#)). Empirically calibrate emissivity from $\geq 10^9 M_{\odot}/h$ haloes to match Ly- α forest opacity, apply to hydro runs;
- Sets of paired uniform UVB and RT runs for isolating effect of **inhomogeneous reionisation** (including pressure smoothing);
- Significant increase in number of models (**375 simulations**, multiple mass resolutions/box sizes, thermal histories, cosmology, WDM, runs to $z=0$);

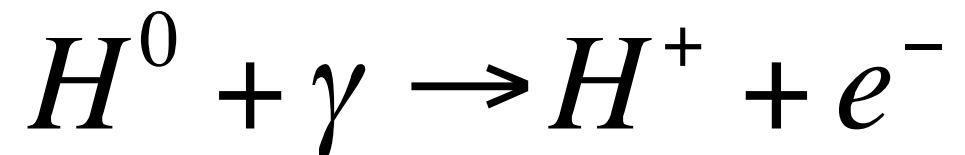


[Puchwein et al. \(2023\)](#)

Photo-heating by UVB



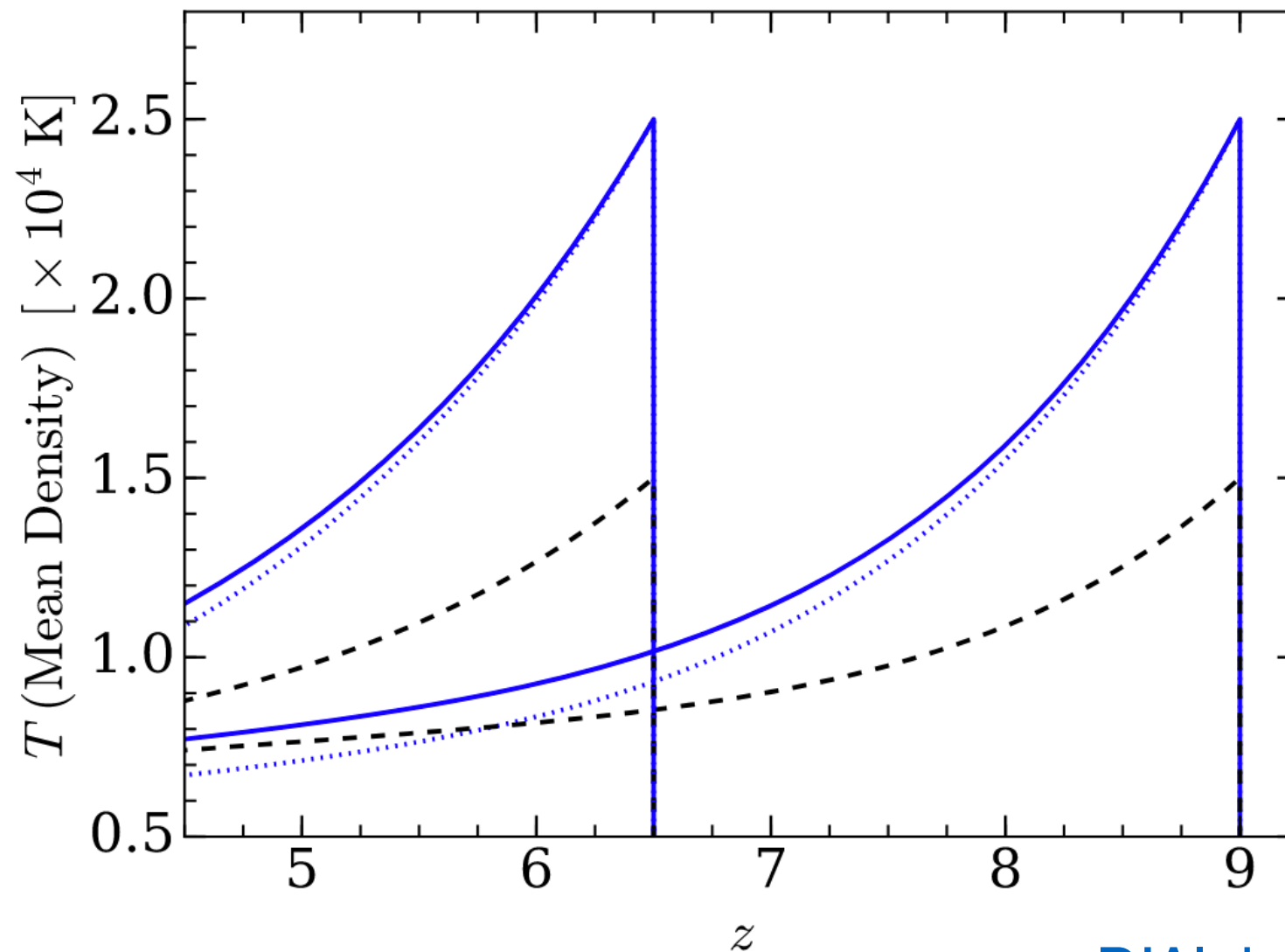
$E > 13.6 \text{ eV}$ photon



($E > 54.4 \text{ eV}$ photon for $\text{He}^+ + \gamma \rightarrow \text{He}^{2+} + e^-$)

Ejected photo-electrons interact with neutrals via scattering and raise temperature of the residual neutral hydrogen.

Photo-heating by UVB

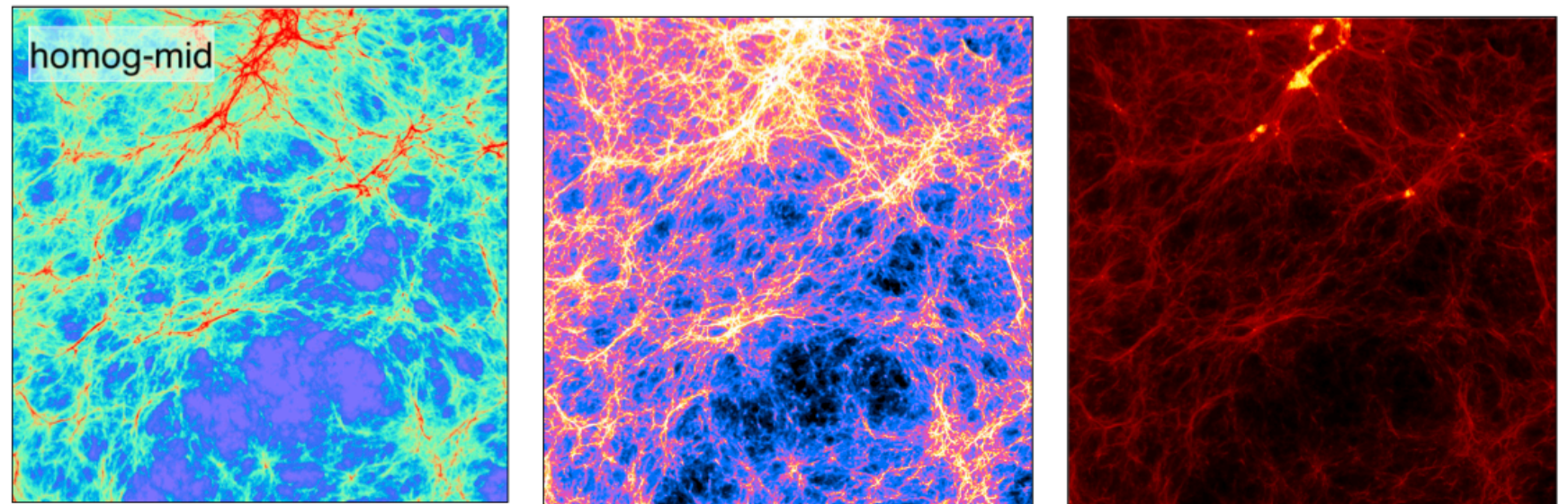


D'Aloisio et al. (2019)

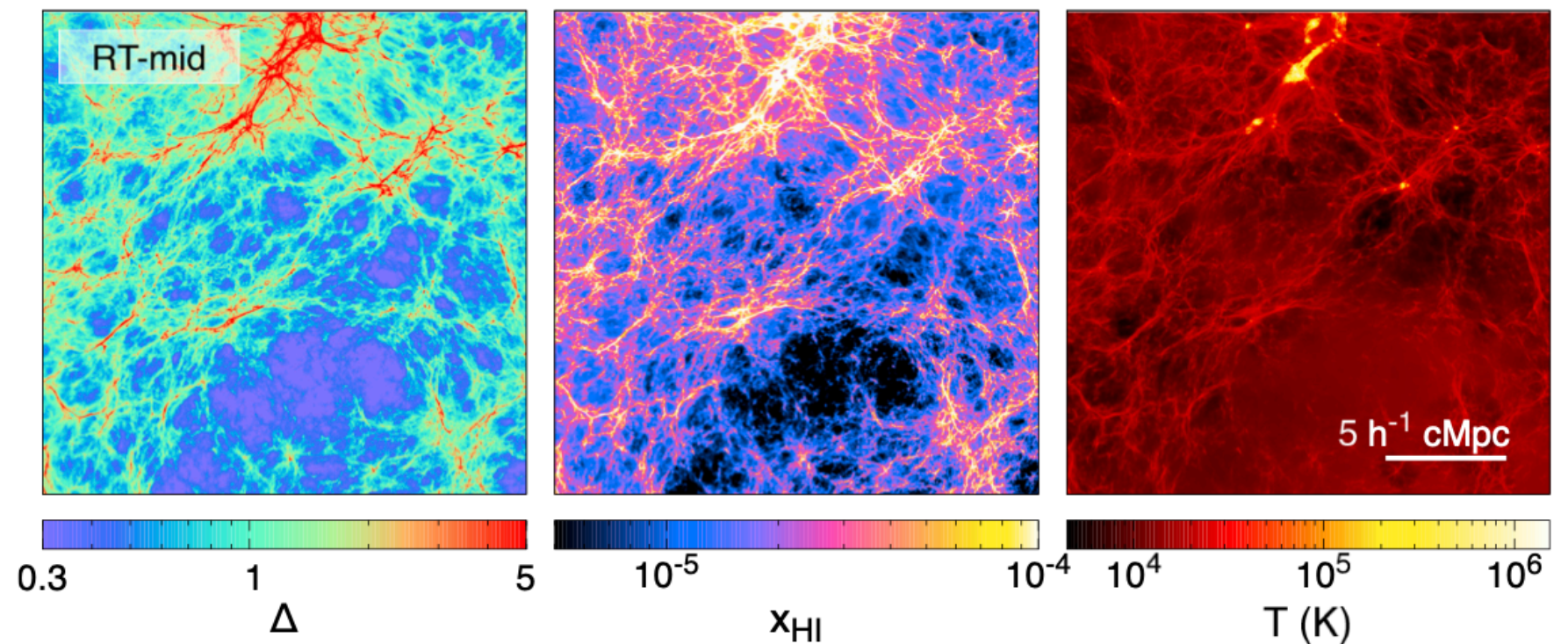
Gas at $\Delta \sim 1$ cools predictably, primarily by adiabatic cooling on timescale $\sim H^{-1}(z)$ (e.g. [Theuns+02](#), [McQuinn & Upton-Sanderbeck 2016](#)).

Patchy photo-heating of IGM

Uniform UVB
at $z=5.4$



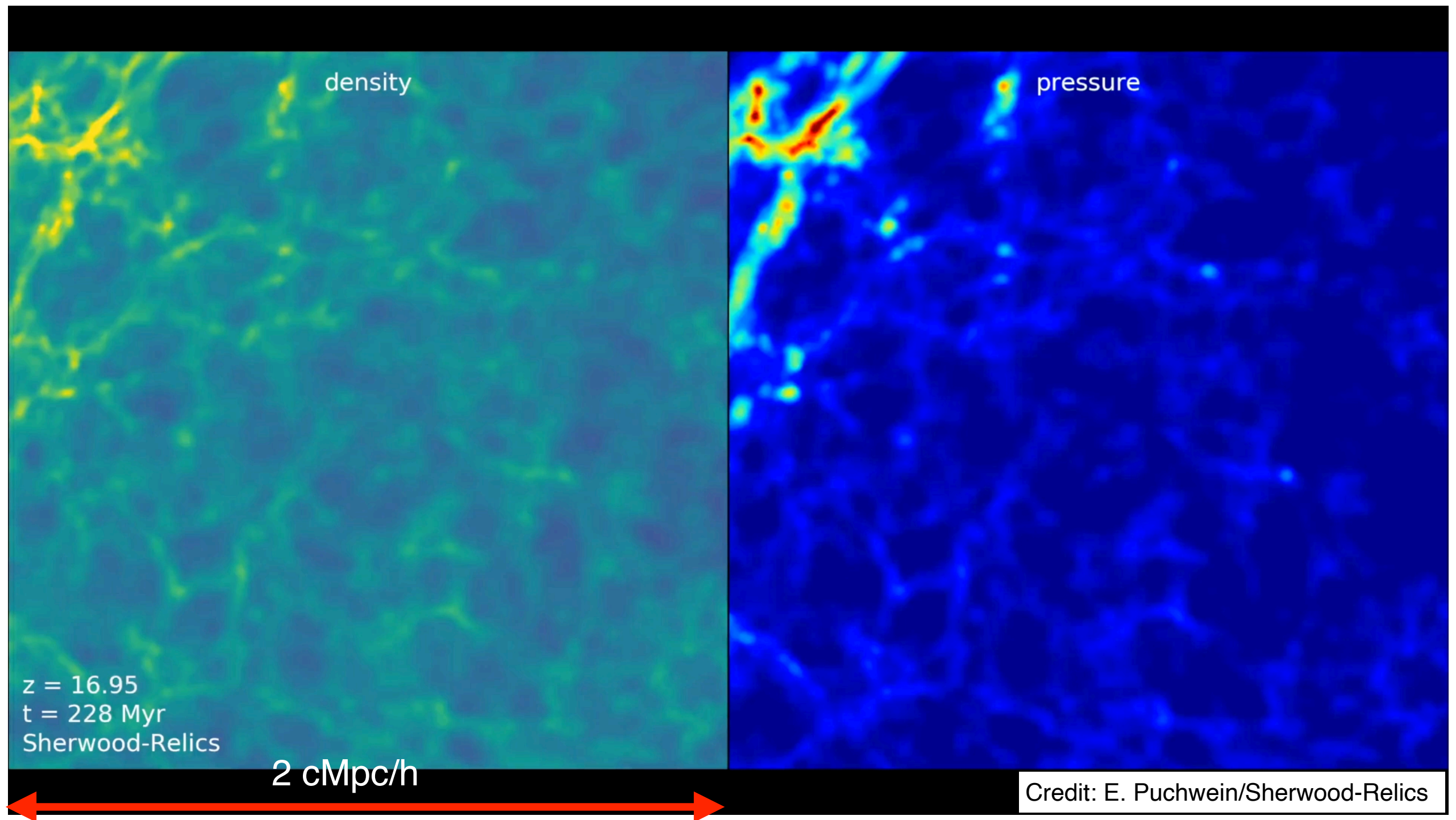
Hydro+RT
model at $z=5.4$



Molaro et al. (2022)

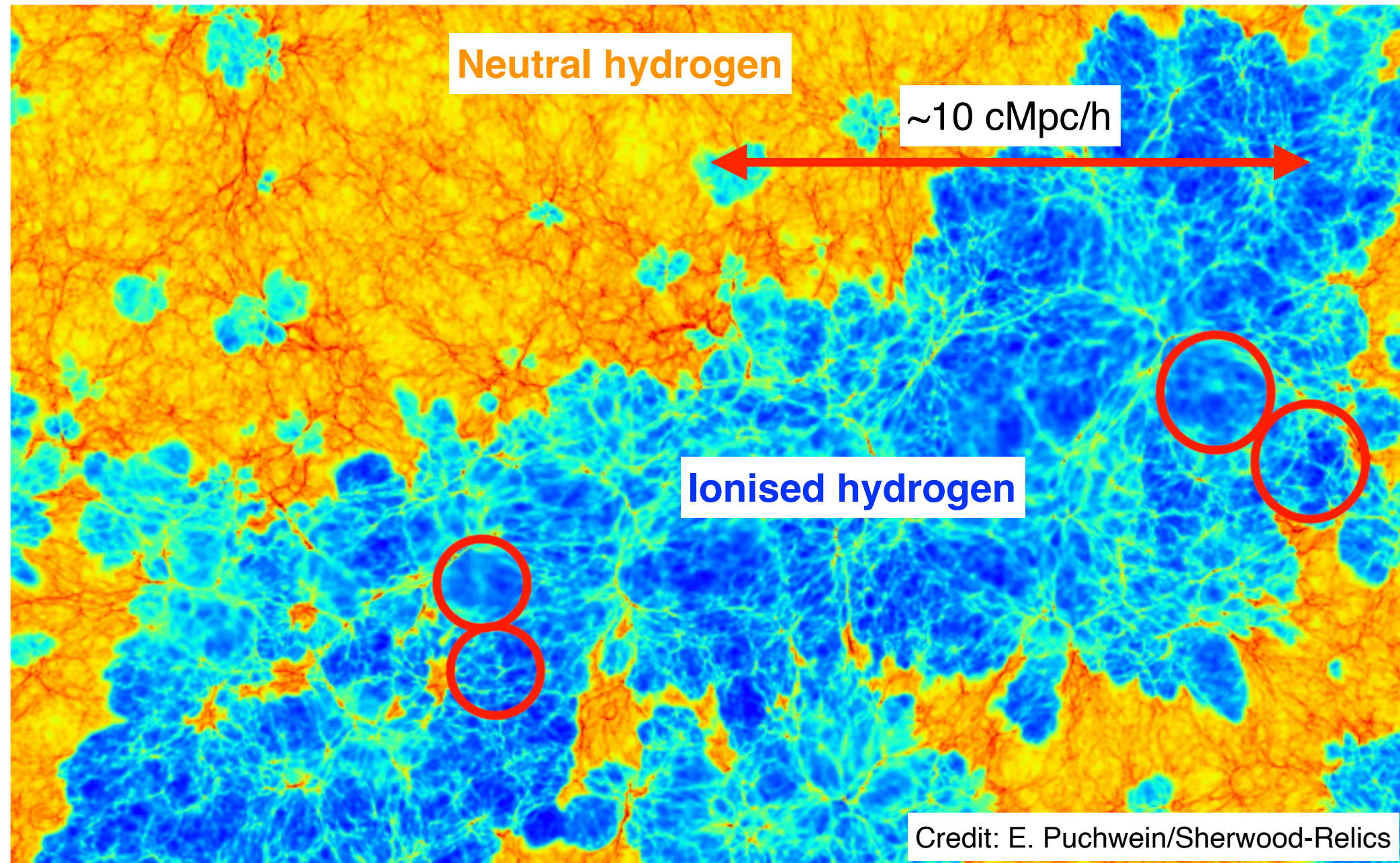
see also Cen+09, Keating+18, Wu+19, Onorbe+19,
D'Aloisio+19, Montero-Camacho & Mao 20

Pressure smoothing



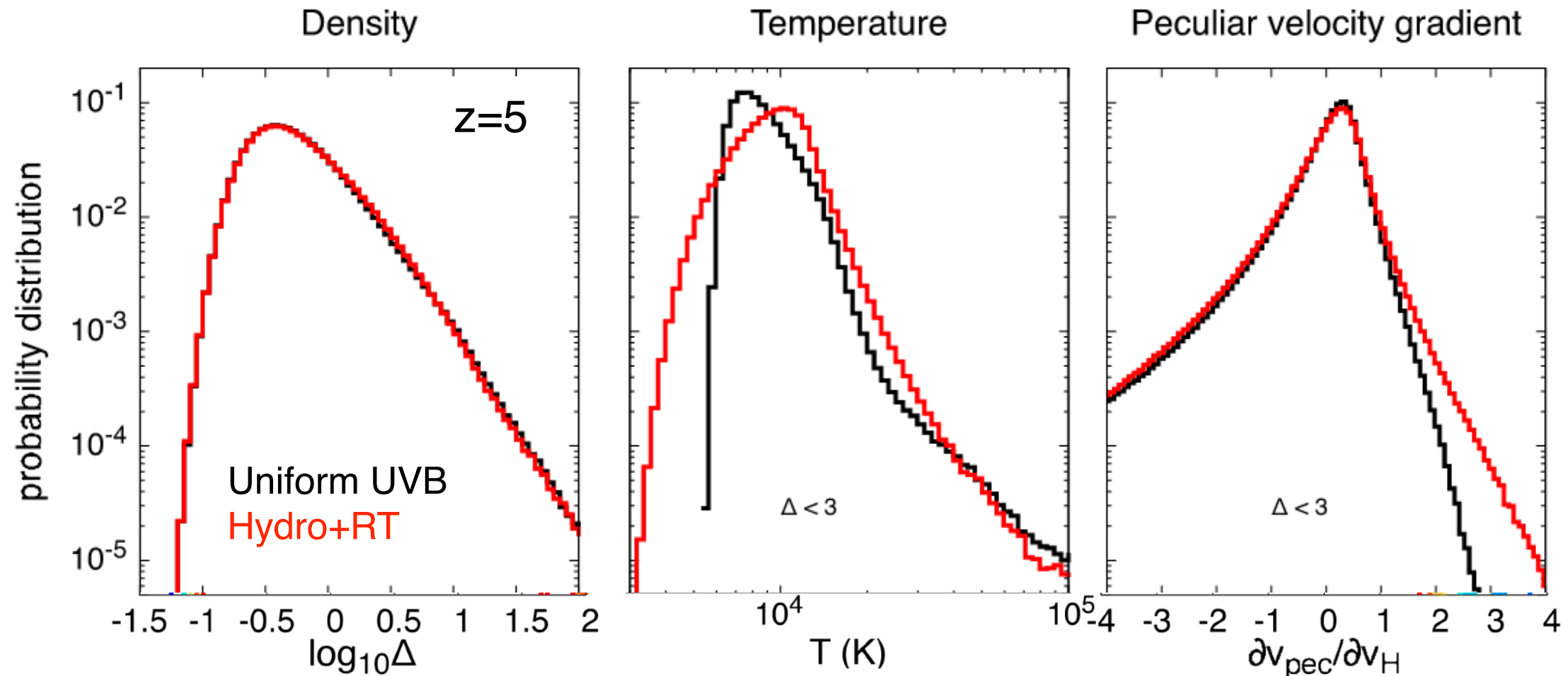
- Radiative transfer is coupled to the hydrodynamical response of the gas;
- Produces spatial variations in the pressure/Jeans smoothing scale of the IGM.

Pressure smoothing at $z=7$



- Radiative transfer is coupled to the hydrodynamical response of the gas;
- Produces spatial variations in the pressure/J Jeans smoothing scale of the IGM.

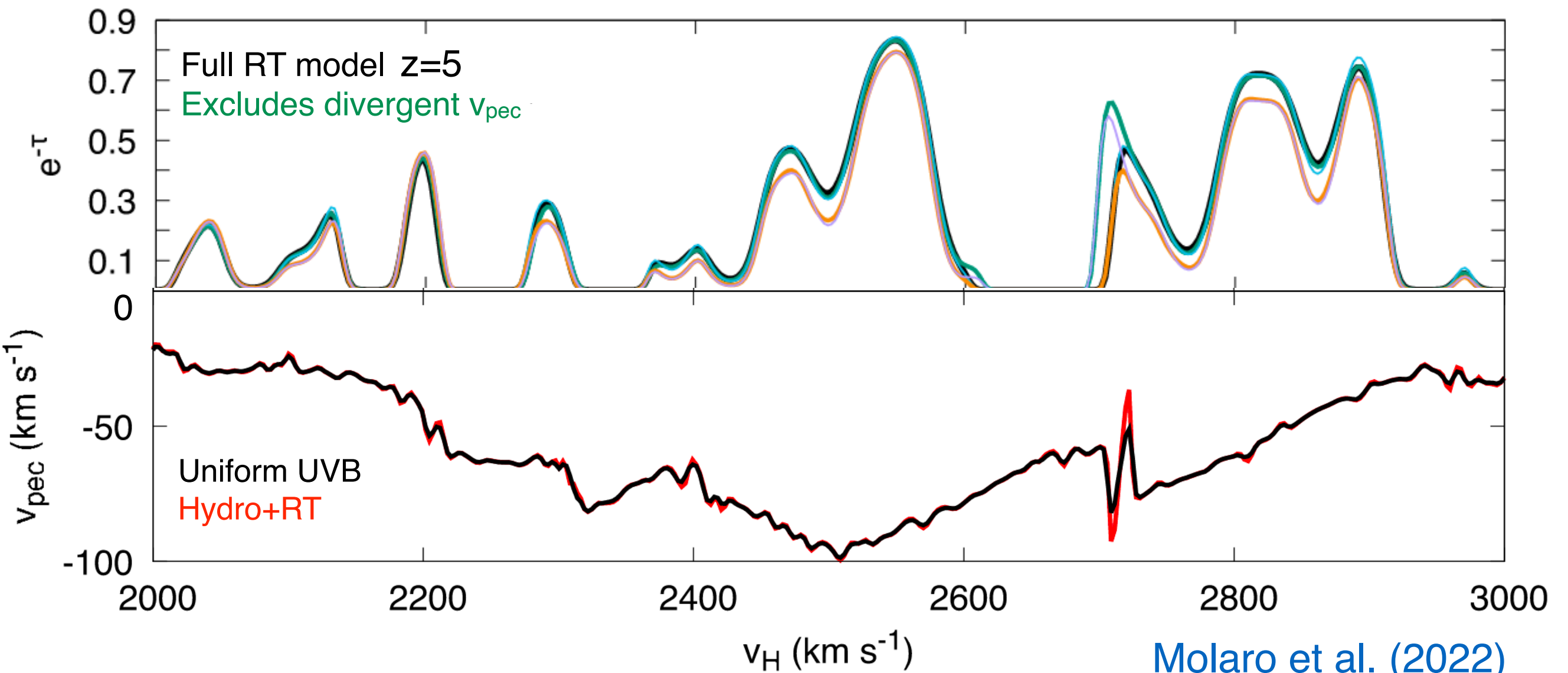
Effect of patchy heating on IGM



Molaro et al. (2022)

- Modest impact on physical gas density distribution by $z=5$;
- Temperature distribution is broader and hotter due to patches of recently reionised gas (cooling time is $\sim H^{-1}(z)$);
- Increase in divergent peculiar velocities, $\partial v_{\text{pec}}/\partial v_H > 0$

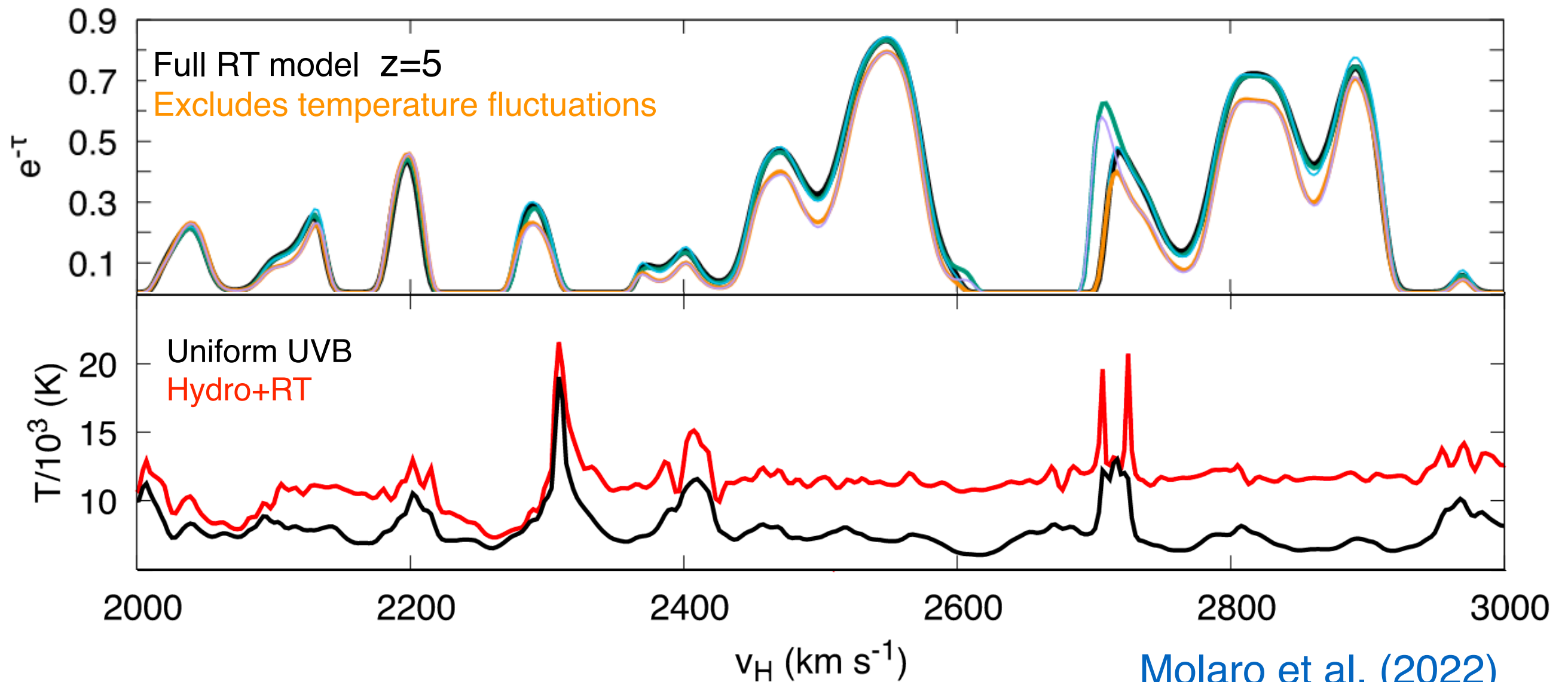
Effect of patchy heating on Ly α



Molaro et al. (2022)

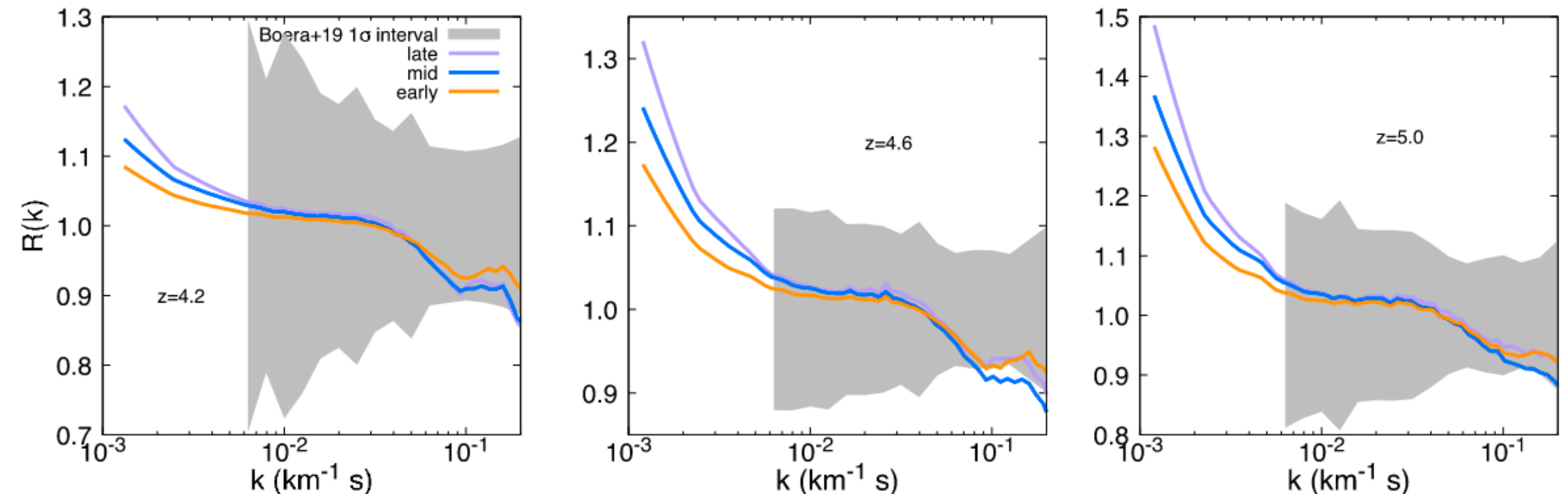
- Divergent peculiar velocities from expanding overdensities broaden Ly α absorption features on small scales;
- Hydrodynamical response to heating needed to capture this.

Effect of patchy heating on Ly α



- Absorption modulated on large scales by fluctuations in gas temperature
Remaining after reionisation, where $\tau_{Ly\alpha} \propto n_{\text{HI}} \propto T^{-0.7}$;
- Predicted by post-processed RT models (e.g. [Keating+18](#)).

Effect on the Ly α 1D power spectrum

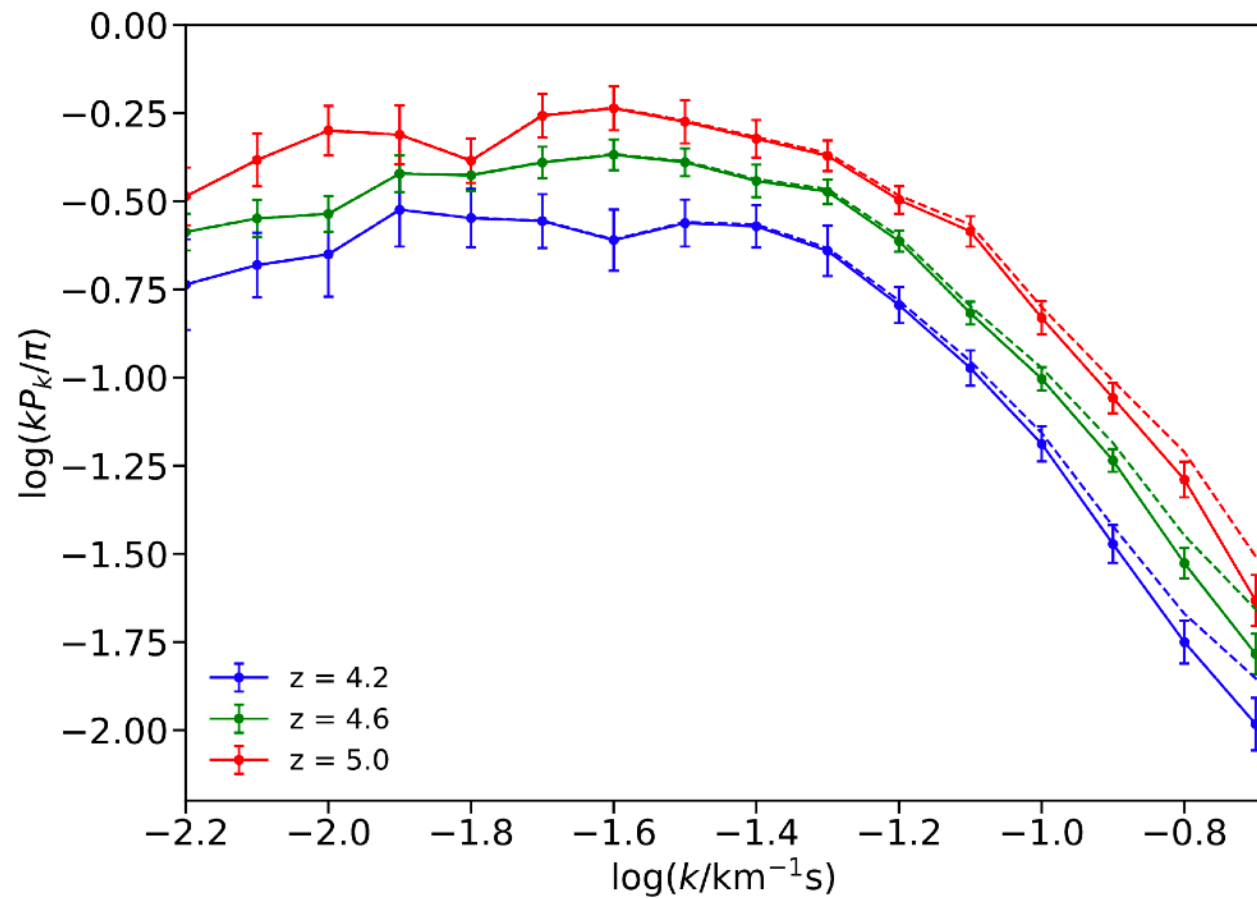


- Enhanced power on large scales, $k < 0.03 \text{ km}^{-1} \text{s}$;
- Suppressed power on small scales, $k > 0.1 \text{ km}^{-1} \text{s}$;
- Effect similar or smaller than 1σ interval on Boera+19 data. Need large scales!

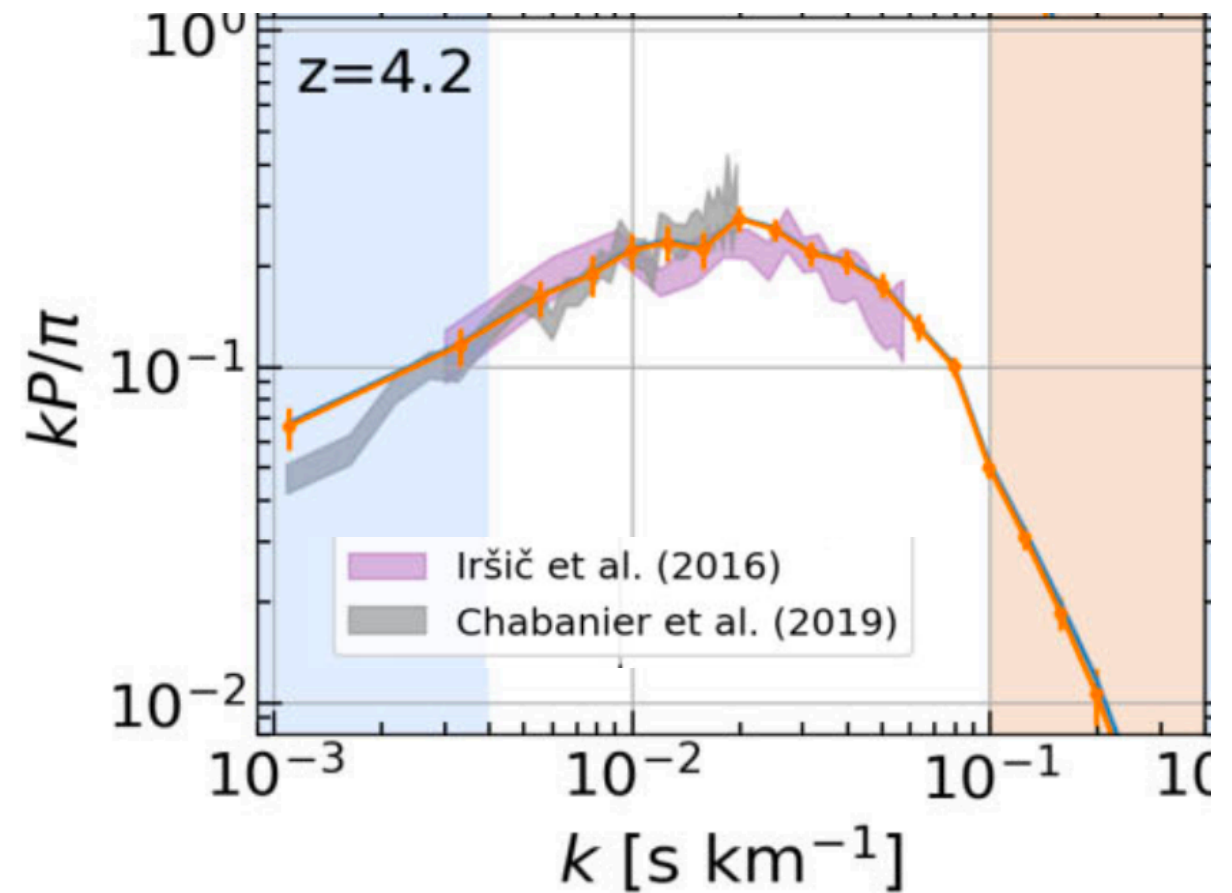
Molaro et al. (2022)

see also Cen+09, Keating+18, Wu+19, Onorbe+19, D'Aloisio+19, Montero-Camacho & Mao 20

The Ly α forest 1D power spectrum



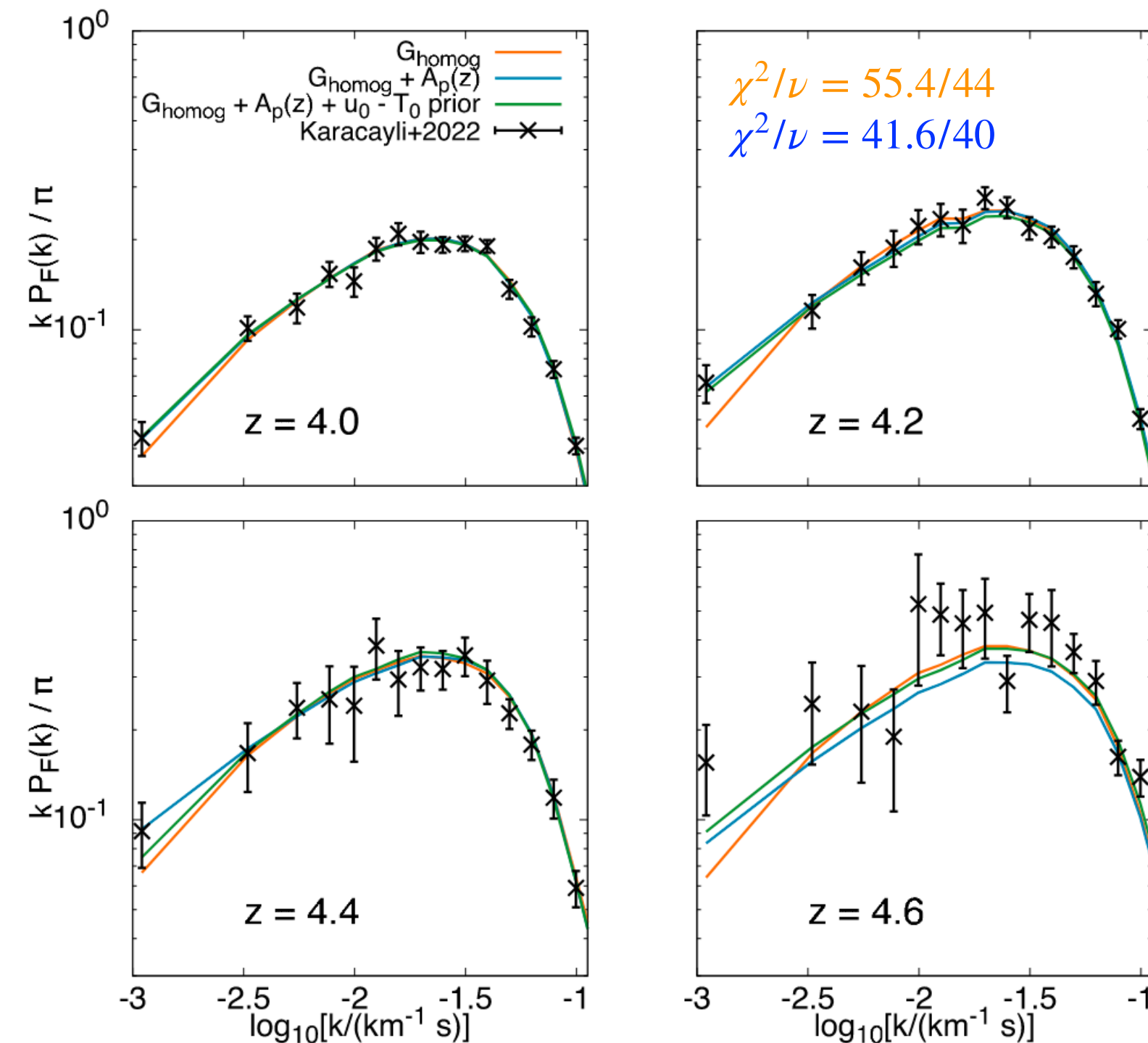
Boera et al. (2019)



Karaçaylı et al. (2022)

- Karaçaylı+22 extend $P(k)$ measurements to $k \sim 10^{-3} \text{ km}^{-1}\text{s}$;
- Based on KODIAQ (O'Meara+17), SQUAD (Murphy+19) and XQ-100 (López+16);
- Data to $z=4.6$ only, but at scales where relic signature of reionisation largest.

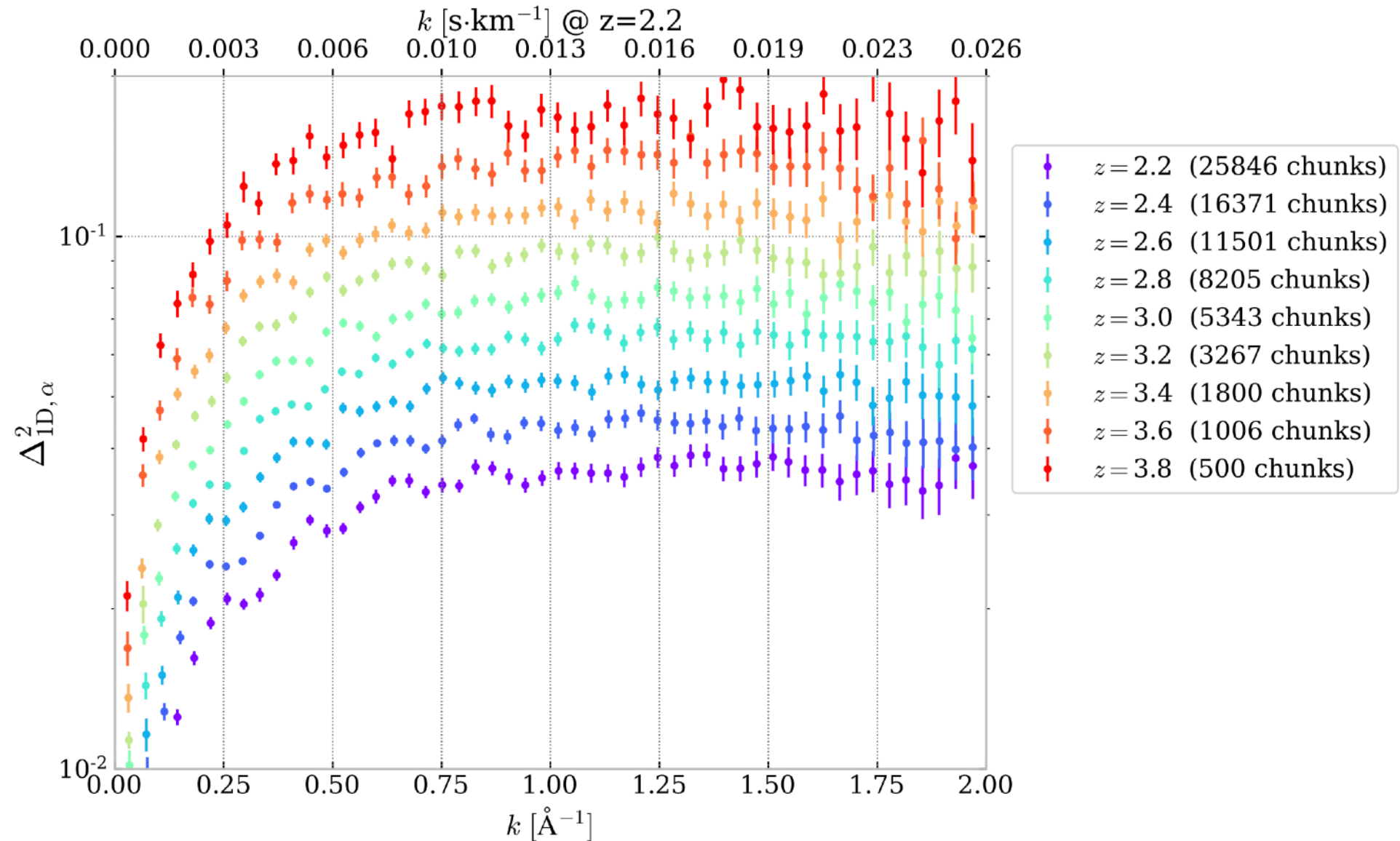
The signature of reionisation?



Molaro et al. (2023)

- Neural network based emulator coupled to Bayesian inference framework;
- 2.7σ preference for an enhancement in power at large scales relative to uniform UVB;
- Possible signature of relic post-reionisation temperature fluctuations? (see also [D'Aloisio+18](#) for an earlier $\sim 2\sigma$ hint at $z=5.4$).

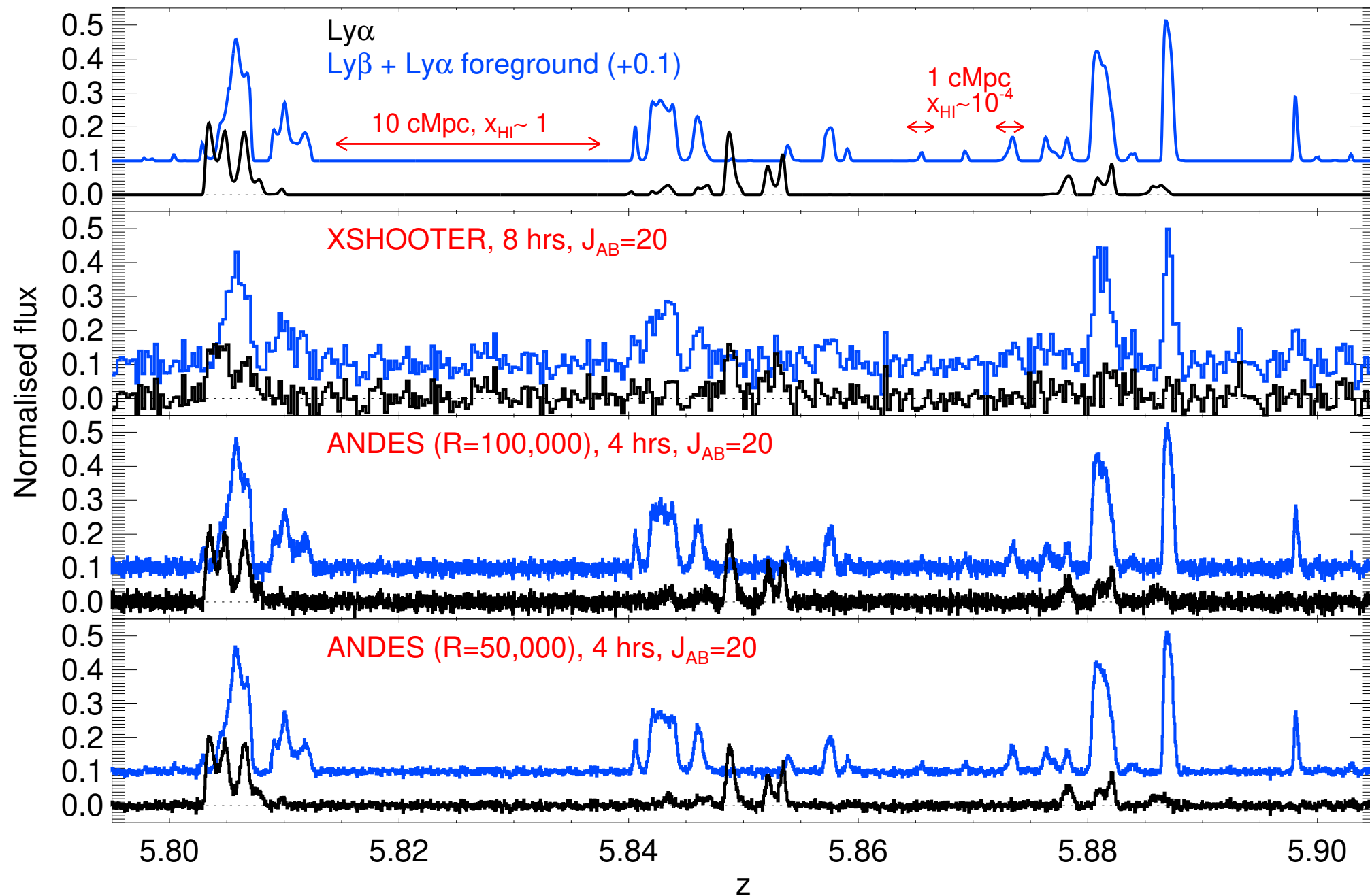
Precision $P(k)$ at large scales



Ravoux et al. (2023), DESI early data release

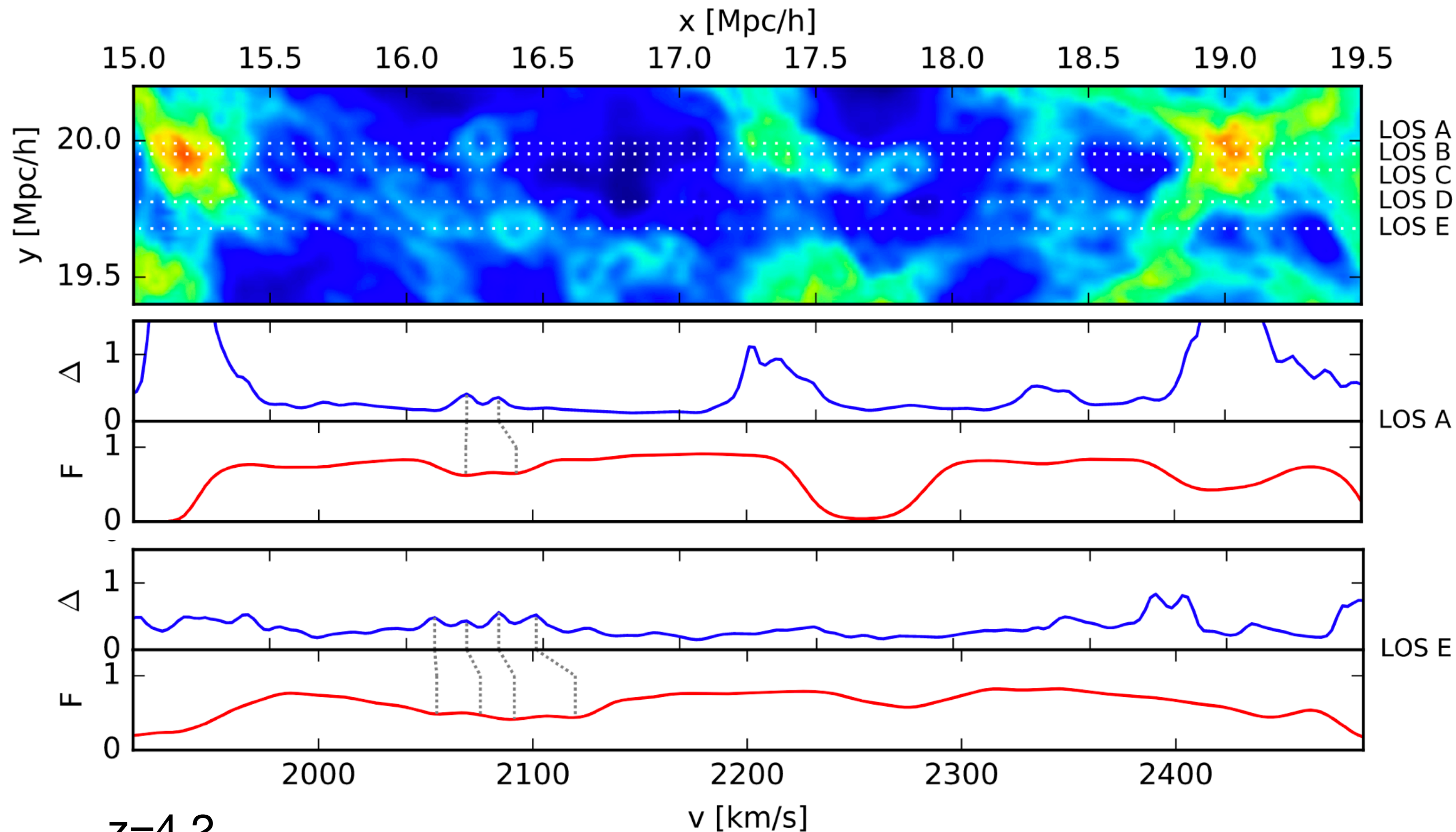
DESI (Vargas-Magana+19) and WEAVE-QSO (Pieri+16) should both measure the 1D $P(k)$ to a few per cent at large scales at $2 < z < 4.5$.

The high- z IGM with ELT/ANDES



ELT/ANDES will be transformative for studying the impact of reionisation on small-scale IGM structure.

Expanding shells in $\text{Ly}\alpha$ absorption



Puchwein et al. (2023)

Summary

- ***Sherwood-relics*** is a large suite of cosmological simulations of the IGM that include radiative transfer coupled to hydrodynamics ([Puchwein+23](#))
- Inhomogeneous reionisation will alter thermal and ionisation state of IGM at $z > 5$. Large-scale temperature fluctuations and divergent peculiar velocities leave characteristic signature in Ly α forest ([Molaro+22](#)).
- Effect small compared to current uncertainties on 1D $P(k)$ measurements (e.g. [Boera+19](#)) at scales used for WDM inference (see also talks by [Iršič](#), [Madau](#)). Good news for existing WDM limits.
- A 2.7σ preference for enhancement in power at $k \sim 10^{-3} \text{ km}^{-1} \text{ s}$ (relative to uniform UVB) in Ly α 1D power spectrum at $4 < z < 4.6$ ([Molaro+23](#)). Signature of (already theoretically predicted) relic post-reionisation temperature fluctuations?
- Future observational progress will be from WEAVE-QSO, DESI, and ELT/ANDES.